

THE NATIONAL RESEARCH PLATFORM

Insights from the 7NRP Conference, and implications for Cal Poly

Brian Spolarich · AVP/CTO · ITS AI Community of Practice · May 2026

AGENDA

What we'll cover

01 What NRP is

Vision, scale, and the headline numbers

03 Who uses it

Researchers, instructors, students, and the LLM service

05 The broader landscape

ACCESS, NAIRR, NERSC, NDP, and what changes next

02 How it works

Kubernetes, Jupyter, GPUs, and the network underneath

04 Where Cal Poly fits

Current usage, sister CSU models, and campus on-ramp

06 Takeaways

Questions for ITS and the AI Community of Practice

Community-owned cyberinfrastructure

"A global-scale Kubernetes cluster for research, education, and public-private partnerships — built and run by the community it serves."

Frank Würthwein · San Diego Supercomputer Center

Research

Freely accessible compute, storage, and GPUs for U.S. researchers and collaborators.

Education

Hosted JupyterHub has become a primary access path for students and instructors.

Partnerships

Vendors, government, and industry contribute hardware, data, and use cases.

The numbers — as of May 2026

129

sites

hosting NRP nodes across 45 U.S. states

1,532

GPUs

shared pool; 74 reserved for LLM serving

32.3K

CPU cores

across 501 registered Kubernetes nodes

22 PB

storage

31 OSDF nodes; capacity to host community data

5,800

users

past year; across 8,000+ projects

151

JupyterHubs

90% of users reach NRP through Jupyter

Source: "State of NRP" keynote, Frank Würthwein, 7NRP, May 2026.

The stack is federated; the front door is simple

User experience

Jupyter notebooks · browser LLM chat · MCP agents

Platform services

JupyterHub · vLLM · Envoy gateway · CILogon · OSDF

Orchestration

Kubernetes "Nautilus" · namespaces · GPU scheduling

Hardware & network

Member-site GPUs/CPUs · Pacific Wave · CENIC · Internet2

Most users meet NRP through a notebook, course link, API token, or shared workspace — not Kubernetes.

What NRP takes off a campus team's plate

Runs hardware from IPMI up

Campuses can contribute racks without building a full local HPC operations team.

Patches the cluster

Cluster-wide security and reliability fixes happen as part of the shared service.

Hosts community data

Useful datasets can be hosted free of charge for the research and education community.

Provides shared services

Identity, GPU scheduling, software catalogs, chat channels, and support pathways.

Notebooks are how most people meet NRP

90%

**of NRP users reach the platform
through Jupyter notebooks.**

5,275 Jupyter users across 151 JupyterHubs.

Zero local setup

Students click a link in Canvas; the environment opens in a browser.

Doing, not watching

Large lectures become executable; students interact with the material.

On-ramp to research

The same environment can scale from a class to graduate and faculty work.

Free access to open-weight LLMs

1. **1** **nrp.ai**

Single sign-on via CILogon → campus IdP

2. **2** **nrp.ai/llmtoken**

OpenAI-style bearer token

3. **3** **API Gateway**

Token check, model routing, usage logging

4. **4** **vLLM GPU pods**

10 frontier open-weight models on call (e.g. Gemma, Qwen3, GPT-OSS)

Why it matters

It gives research and teaching workflows a free, federated alternative to commercial inference services, with usage tied back to campus identity.

The demand signal is already visible

17 B

tokens / week peak

\$300K of inference in the past 3 months

751

users · 276 projects

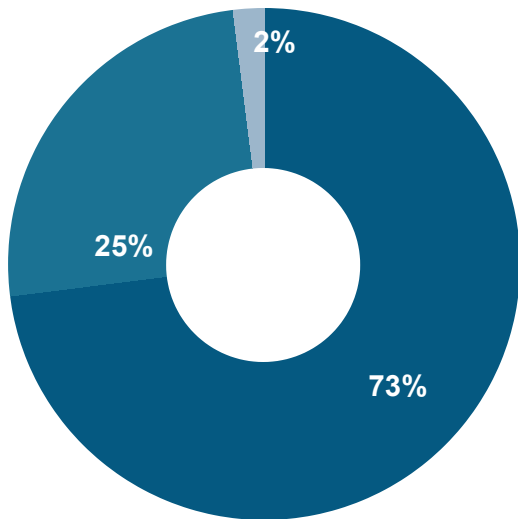
across 35 institutions; growing ~0.5 B tokens/week

72%

of token usage in CA

95% input-heavy → RAG, not chat

Researchers, instructors, students



More non-R1 than R1 institutions

97 non-research-intensive institutions vs. 54 R1s.

Education is the dominant workload

Most usage is classroom Jupyter, often for undergraduates in early programming classes.

Free at point of use

Anyone with a U.S. institutional identity can request access; students need PI approval.

The on-ramp for a new campus

1 Faculty explore

A few faculty open accounts and start using shared resources.

2 Demand builds

Interest crosses the threshold where campus hardware starts to make sense.

3 REN connects

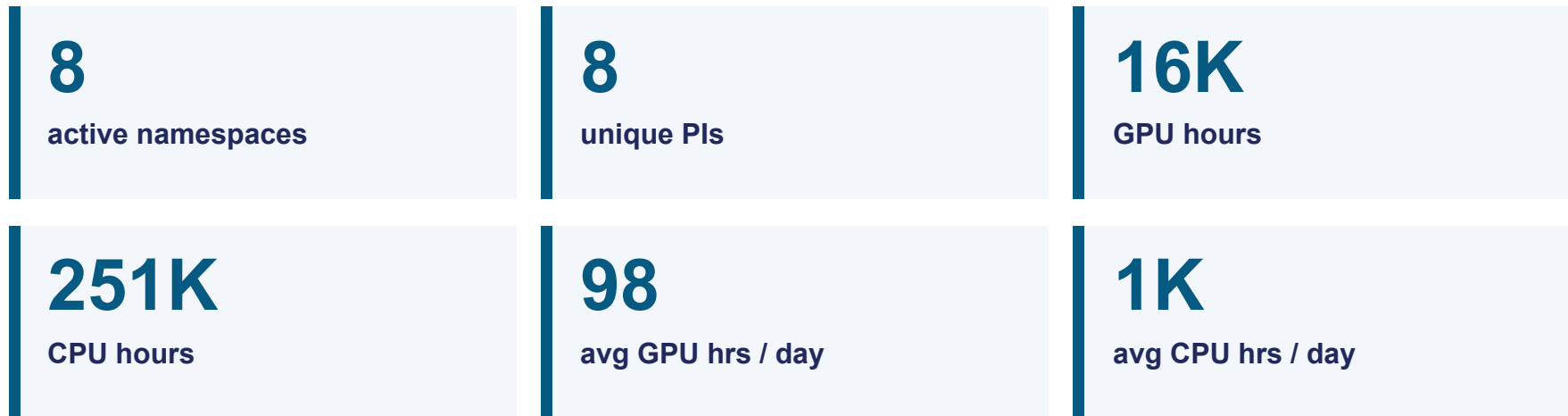
CENIC provisions a Science DMZ, typically 10G+.

4 NRP operates

Campus racks hardware; NRP runs it from IPMI up.

Cal Poly already has a Science DMZ in place.

What's already happening at Cal Poly



Active NRP usage over the past 12 months.

Source: NRP Grafana usage dashboard, Cal Poly namespaces, May 2026.

Lauren Chan — Cal Poly SLO

Evolutionary Biology & Conservation Computational Genomics

Research featured in "Whirlwind Tour of Science on NRP," Larry Smarr, 7NRP 2026.

892 GPU hours and 27.3K CPU hours to date

Reference-genome decontamination on high-memory nodes

Variant calling across full genomes for 100 individuals

Post-calling filtering and downstream analysis

Three operating models to learn from

SDSU

VERNE + TIDE

Instructional cluster, JupyterHub front-end, student-assistant support model.

Demand creation is the hard part.

CSU SAN BERNARDINO

BOT model

Build, Operate, Transfer for partners; preconfigured JupyterHubs.

Stage resources ahead of demand.

CSU FULLERTON

Faculty enablement

IT + faculty + Academic Affairs; modules co-designed with instructional designers.

Jupyter is not just for STEM.

Science examples, part 1

AGRICULTURE

AIIRA & Iron Horse Vineyard

InsectNet/WeedNet foundation models
and a connected vineyard testbed.

WILDFIRE

BurnPro3D

Operational 3D fire and smoke
modeling for prescribed burns.

CLIMATE

JAX-based climate emulators

Portable models running on
CPU/GPU/TPU, bridging observations
and simulation.

Science examples, part 2

HEALTH

Brain MRI for Alzheimer's

CSU Fullerton: 4,500 GPU hours across 1,200 participants.

MATERIALS

Quantum molecular dynamics

CSUSB: VASP simulations using custom JupyterHubs and 50K GPU hours/year.

GENOMICS

Deep regulatory genomics

Stanford and Cal Poly work decoding gene regulation on the same platform.

UCSD shows what digital assets can become

30,000

**students per year on UCSD
DataHub, across 30 departments.**

2017

Free compute for instructors

Shared notebook infrastructure free at point of use.

2024

AI tutors per class

Personalized tutors across multiple large courses.

2026

Free AI for instructors

Open-weight LLMs at the same scale.

The California transfer pipeline is the scale problem

K-12

2M+

public high schoolers

CCC

2.2M

students across 116 colleges

CSU

480K

students across 23 campuses

UC

440K

students across 10 campuses

30% of U.S. transfer students come from California community colleges. If CSUs and UCs deploy digital assets but CCCs cannot match them, the pipeline breaks.

Where NRP sits in U.S. research computing

	NRP	ACCESS	NAIRR	NERSC	NDP
Purpose	Research + education	NSF compute allocations	AI research resource	DOE leadership computing	Federated AI-ready data
Access	Open at point of use	Proposal-based	3-page request	DOE-aligned PI allocation	Federation; API + endpoints
Scale	5,800 users · 8K+ projects	12K users / yr	700+ projects · 6K+ students	11K researchers / yr	Data + workspaces
What you get	K8s + Jupyter + LLMs	Many systems	Academic + industry GPUs	Leadership systems	Catalogs, classrooms

Agents, data, and an AI-ready ecosystem

Agentic AI on trusted infrastructure

Agents that discover data, assemble workflows, and act on scientific systems.

Federation of NRP + NDP

Endpoints anywhere, connected to a national catalog.
Workflow is not teamflow.

Edge to HPC continuum

Sensors, drones, inference, JupyterHubs, and leadership-class systems in one ecosystem.

Sustaining the pilots

NAIRR moves toward at-scale; NRP explores longer-term funding models.

KEY TAKEAWAYS

What I'd like you to walk away with

NRP is real, free, and growing fast.

129 sites · 1,532 GPUs · 5,800 users · 90% through Jupyter.

Kubernetes is the substrate; Jupyter is the front door.

The interesting layer is the notebook and service experience.

Research and education are now side by side.

The same platform can serve classrooms, labs, and AI workflows.

California is central — and Cal Poly fits the profile.

CENIC is the network; sister CSUs are already running playbooks.

The boundaries are dissolving.

This crosses research, education, IT, data, and AI strategy.

RESOURCES

Where to learn more

nrp.ai

Platform access, namespaces, docs

nrp.ai/llmtoken

Generate an LLM API token

nationaldatapatform.org

NDP catalog, classrooms, federation docs

nairrpilot.org

NAIRR resource request form

dash.nrp-nautilus.io

NRP dashboard and Grafana metrics

data8.org / data6

Open intro data science curriculum

Want to talk further? brianspo@calpoly.edu · Slides and notes available on request.

Q U E S T I O N S

What's catching your attention?

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